

PRODUCTION AI ENGINEERING · O1 + O2

EVIDENCE CHECK

Evidence Check: every claim, traced to its proof

Each claim of the O1 + O2 editions: the scenario, the mechanism, the checks behind it, what the run observed, and the class a reader should give it.

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O1 + O2 · Production AI Engineering

9 min read · Published 2026-10-08 · Evidence Check

How to read this

Every claim is stated without its numbers; the numbers are the checks' observed values, substituted from the published run. A check compares a recorded fact with a preregistered value or with the other arm. **SUPPORTED** means every check held; **QUALIFIED** means it held, with a limitation the run observed (a check marked LIMITATION OBSERVED); **NEGATIVE CONTROL** means a property broke on purpose when its safeguard was removed; **ARGUED** and **NOT TESTED** are reasoned or out of scope.

All values are **simulation units** (simulated milliseconds, cost units). They show that a mechanism behaves as claimed under a declared workload; they are not benchmarks of any real provider, model or platform.

The proof pack: 11 experiments, 65 checks (63 pass, 1 fail as a recorded limitation, 1 expected failure), 14 claims. Replay EXACT: 99 of 99 files byte-identical. The same mapping, machine-readable: `results/claim-evidence.json`.

OPS-C01 · SUPPORTED

Capacity should be enforced before unlimited work enters the runtime: bounding work in system kept the surge out of the runtime, and open admission turned timeouts into retries and an unbounded queue.

Scenario: OPS-E1 · Admission before execution. Black Friday surge: support requests at a multiple of capacity for 120 s, then the evening; same clients in both arms (they give up after 20 s and retry).

Mechanism (the only variable): Admission: open (unbounded queue) vs bounded work in system with a capacity error and Retry-After.

Check	What must hold	Observed	Result
OPS-E1-C01	Bounded admission: work in system never exceeded slots + queue bound	128	PASS
OPS-E1-C02	Open admission: the queue grew past the queue bound	5,759	PASS
OPS-E1-C03	Retry amplification is higher with open admission	2.8	PASS
OPS-E1-C04	Goodput with bounded admission is at least goodput with open admission	45.5	PASS
OPS-E1-C05	Bounded admission: p99 queue delay of admitted attempts is below the client timeout	14,027	PASS
OPS-E1-C06	Bounded admission refuses some requests explicitly (the cost of control)	1,405	PASS
OPS-E1-C07	Recomputed from the raw rows: work in system under bounded admission stays within the bound	128	PASS

Raw evidence: `ops_poc/runs/2026-10-08-recorded/` (see each check's `evidence` in `evidence/runs/2026-10-08-recorded/checks.jsonl`).

OPS-C02 · SUPPORTED

Multi-tenant agent platforms need tenant-aware resource controls, not only global limits: under one global FIFO the finance run starved support; with weighted fair scheduling and a bulkhead, support was served and the batch still completed, later.

Scenario: OPS-E2 · Tenant fairness: the noisy neighbour is your own finance team. Support at a steady rate; finance submits its reconciliation run in ten seconds; same 32 slots.

Mechanism (the only variable): Scheduling: one global FIFO vs weighted fair scheduling with a finance bulkhead.

Check	What must hold	Observed	Result
OPS-E2-C01	Fair scheduling: at least 99 % of support requests served within the client timeout	100	PASS
OPS-E2-C02	Global FIFO serves fewer support requests within the timeout	3.6	PASS
OPS-E2-C03	Fair scheduling: finance never ran more than its bulkhead at once	16	PASS
OPS-E2-C04	Fair scheduling: every finance item still completed	3,000	PASS
OPS-E2-C05	Fair scheduling: the finance batch finished within its deadline	374.9	PASS
OPS-E2-C06	The batch took longer under fair scheduling (the cost of control)	374.9	PASS

Raw evidence: `ops_poc/runs/2026-10-08-recorded/` (see each check's `evidence` in `evidence/runs/2026-10-08-recorded/checks.jsonl`).

OPS-C03 · SUPPORTED

Concurrency must be bounded independently of incoming request volume: unbounded concurrency slowed every workflow and spent most of its cost on attempts nobody was waiting for; slots, a bounded queue and deadline-aware dequeue kept goodput up.

Scenario: OPS-E3 · Bounded concurrency, backpressure and deadlines. Overload at 1.5x capacity for 60 s, then light load; same clients.

Mechanism (the only variable): Unbounded concurrency vs slots + bounded queue + deadline-aware dequeue + cancellation.

Check	What must hold	Observed	Result
OPS-E3-C01	Bounded: running attempts never exceeded the slots	32	PASS
OPS-E3-C02	Unbounded: running attempts exceeded the runtime's capacity	1,227	PASS
OPS-E3-C03	Deadline-aware dequeue: no attempt started after its client's deadline	0	PASS
OPS-E3-C04	Less cost spent on attempts nobody was waiting for	7.8	PASS
OPS-E3-C05	Higher goodput with the controls	84.4	PASS
OPS-E3-C06	Recomputed from the raw rows: running attempts never exceeded the slots	32	PASS

Raw evidence: `ops_poc/runs/2026-10-08-recorded/` (see each check's `evidence` in `evidence/runs/2026-10-08-recorded/checks.jsonl`).

OPS-C04 · QUALIFIED

Production agents need explicit execution budgets, propagated to child workflows: the envelope stopped a runaway loop and bounded a coordinator with sub-agents, where per-agent budgets did not; calibrated on a development sample, it also stopped some legitimate workflows of the blind sample.

Scenario: OPS-E4 · Workflow resource envelopes, propagated to children. A refund whose payment never settles; a coordinator with three sub-agents, one stuck, re-delegated; 350 calibrated normal workflows.

Mechanism (the only variable): Budgets: none vs per-agent vs a propagated envelope.

Check	What must hold	Observed	Result
OPS-E4-C01	Envelope: the runaway refund stopped with BUDGET_EXCEEDED	BUDGET_EXCEEDED	PASS
OPS-E4-C02	No budget: the runaway ran until the simulator's harness guard	400	PASS
OPS-E4-C03	Propagated envelope: the coordinator's model calls (children included) stayed within its envelope	9	PASS
OPS-E4-C04	Per-agent budgets: the coordinator's total model calls exceeded its envelope	11	PASS
OPS-E4-C05	The calibrated envelope stopped no legitimate workflow of the blind sample	4	LIMITATION OBSERVED

Raw evidence: `ops_poc/runs/2026-10-08-recorded/` (see each check's `evidence` in `evidence/runs/2026-10-08-recorded/checks.jsonl`).

OPS-C05 · SUPPORTED

Model routing should optimise within a declared quality/policy contract, not cost alone: inside the contract, routing cost less per successful workflow than all-large with no violation, deferring explicitly during a throttle; optimising cost alone (all-small, any fallback) left the contract.

Note: Synthetic eligibility contract and success table: no statement about the quality of any real model.

Scenario: OPS-E5 · Routing inside an eligibility contract. 800 workflows (support mix + finance), the EU private deployment throttled for 30 s.

Mechanism (the only variable): All-large vs all-small vs routed vs routed with any fallback.

Check	What must hold	Observed	Result
OPS-E5-C01	Routed costs less per successful workflow than all-large	10	PASS
OPS-E5-C02	Routed: no model call below the task's capability tier	0	PASS
OPS-E5-C03	Routed: no payment data on a profile not allowed to process it, throttle included	0	PASS
OPS-E5-C04	Routed success rate within 1 percentage point of all-large (or better)	2.7	PASS
OPS-E5-C05	All-small costs less per request than routed	2.5	PASS
OPS-E5-C06	All-small succeeds less often than routed	98	PASS
OPS-E5-C07	Routed with any fallback put payment data on a global profile during the throttle	18	PASS
OPS-E5-C08	Routed deferred explicitly during the throttle instead of crossing the policy	7	PASS

Raw evidence: `ops_poc/runs/2026-10-08-recorded/` (see each check's `evidence` in `evidence/runs/2026-10-08-recorded/checks.jsonl`).

OPS-C06 · SUPPORTED

Context and retrieval need resource controls like model execution: bounded retrieval kept every declared required evidence id with far fewer tokens, and a cache keyed by tenant, principal scope and versions hit without leaking or going stale, unlike a query-text cache.

Scenario: OPS-E6 · Context budgets and cache scope. 12 fixture queries with declared required evidence ids; a 24-lookup cache sequence with a knowledge and a policy version change.

Mechanism (the only variable): Naive vs bounded retrieval; query-text vs scoped cache key.

Check	What must hold	Observed	Result
OPS-E6-C01	Bounded retrieval kept every required evidence id of every query	0	PASS
OPS-E6-C02	Bounded retrieval used fewer context tokens per query	5,982	PASS
OPS-E6-C03	Scoped cache: no personal answer served to another customer	0	PASS
OPS-E6-C04	Scoped cache: no entry from an older knowledge or policy version served	0	PASS
OPS-E6-C05	Query-text cache served a personal answer to another customer	5	PASS
OPS-E6-C06	Query-text cache served stale entries after a version change	5	PASS
OPS-E6-C07	The scoped cache still hit on shared questions	7	PASS

Raw evidence: `ops_poc/runs/2026-10-08-recorded/` (see each check's evidence in `evidence/runs/2026-10-08-recorded/checks.jsonl`).

OPS-C07 · SUPPORTED

Tool capacity must be governed independently of model capacity: the gateway held the payments API inside its capacity with no 503s and far fewer attempts; alone on a FIFO queue it moved the wait into the runtime's slots, and composed with fair scheduling it served support too.

Scenario: OPS-E7 · A tool gateway in front of a constrained downstream. Payments-status API (capacity 8, connection limit 16, 20 rps) behind a 64-slot runtime; a finance run and support traffic.

Mechanism (the only variable): Direct calls with immediate retries vs the gateway vs the gateway composed with fair scheduling.

Check	What must hold	Observed	Result
OPS-E7-C01	Gateway: calls in flight never exceeded the API's capacity	8	PASS
OPS-E7-C02	Gateway: the API returned no 503	0	PASS
OPS-E7-C03	Direct calls reached the API's connection limit	16	PASS
OPS-E7-C04	More downstream attempts per logical call with direct calls	2.8	PASS
OPS-E7-C05	At least as many workflows completed with the gateway	3,584	PASS
OPS-E7-C06	Gateway on one FIFO queue: support goodput fell below direct calls (the wait moved into the runtime's slots)	5.3	PASS
OPS-E7-C07	Gateway + fair scheduling: support goodput at least that of direct calls	100	PASS
OPS-E7-C08	Gateway + fair scheduling: the API returned no 503	0	PASS
OPS-E7-C09	Gateway + fair scheduling: every finance item completed	1,500	PASS

Raw evidence: `ops_poc/runs/2026-10-08-recorded/` (see each check's `evidence` in `evidence/runs/2026-10-08-recorded/checks.jsonl`).

OPS-C08 · SUPPORTED

Behaviourally relevant non-code changes must create a distinguishable agent release: six single-artifact changes produced six new release ids under one unchanged image digest, and every telemetry row is attributable to its release. (The behaviour a tool description causes is declared in the scripted agent.)

Scenario: OPS-E8 · The behavioural release. R41 and six candidates, each changing one artifact; the same 300-workflow Cyber Monday sample replayed under each.

Mechanism (the only variable): One artifact per candidate.

Check	What must hold	Observed	Result
OPS-E8-C01	The image digest is identical across the seven releases	1	PASS
OPS-E8-C02	Seven releases, seven release ids	7	PASS
OPS-E8-C03	Every candidate differs from R41 in exactly one artifact	1	PASS
OPS-E8-C04	The declared behavioural model is wired: a tool description alone changes tool calls per workflow on the same workload (scripted, so it cannot fail; reclassified after the freeze, DEVIATIONS.md A2)	1.9	PASS
OPS-E8-C05	Every telemetry row of the run carries a release id	58,573	PASS

Raw evidence: `ops_poc/runs/2026-10-08-recorded/` (see each check's `evidence` in `evidence/runs/2026-10-08-recorded/checks.jsonl`).

OPS-C09 · SUPPORTED

Behavioural releases should pass machine-checkable gates before receiving production traffic: three candidates with good averages and one broken invariant each were blocked and granted no traffic.

Scenario: OPS-E9 · Offline evals and invariant gates. A 40-case offline suite (historical, adversarial, synthetic) under R41 and five candidates.

Mechanism (the only variable): The candidate release.

Check	What must hold	Observed	Result
OPS-E9-C01	The gate blocked exactly R42-b, R42-c and R42-d	R42-b, R42-c, R42-d	PASS
OPS-E9-C02	No traffic was granted to a blocked candidate	0	PASS
OPS-E9-C03	R42-b's average task success met the threshold (the average looked fine)	1	PASS
OPS-E9-C04	R42-a and R42-e passed the gate	R42-a, R42-e	PASS

Raw evidence: `ops_poc/runs/2026-10-08-recorded/` (see each check's `evidence` in `evidence/runs/2026-10-08-recorded/checks.jsonl`).

OPS-C10 · SUPPORTED

Agent releases require behavioural canaries and rollback on operational evidence, not process health: a release that passed the offline gate and kept its error rate rolled back on tool-call amplification; a clean one was promoted; a raw-average comparison would have rolled the clean one back; the rollback did not undo replies already sent.

Scenario: OPS-E10 · A behavioural canary, promotion and rollback. Cyber Monday support traffic; 10 / 50 / 100 % stages; windows of 60 candidate workflows; frozen guardrails.

Mechanism (the only variable): The candidate (R42-a, R42-e) and the comparison (mix-adjusted, raw).

Check	What must hold	Observed	Result
OPS-E10-C01	The canary rolled R42-a back	ROLLBACK	PASS
OPS-E10-C02	R42-a's error rate stayed within 1 pp of R41 in every window (healthy by status code)	0	PASS
OPS-E10-C03	After the rollback no request reached R42-a	0	PASS
OPS-E10-C04	The canary promoted the clean R42-e	PROMOTE	PASS
OPS-E10-C05	Replies sent by R42-a's canary workflows before the rollback	60	PASS
OPS-E10-C06	None of them was undone by the rollback	0	PASS
OPS-E10-C07	A raw-average comparison rolled the clean R42-e back	ROLLBACK	PASS

Raw evidence: ops_poc/runs/2026-10-08-recorded/ (see each check's evidence in evidence/runs/2026-10-08-recorded/checks.jsonl).

OPS-C11 · NEGATIVE CONTROL

The proof can fail: with the admission bound removed, the E1 invariant check breaks.

Scenario: OPS-NC · Negative control: admission removed. E1's controlled arm, identical except that the admission bound is removed.

Mechanism (the only variable): The admission bound.

Check	What must hold	Observed	Result
OPS-NC-C01	Without the bound, work in system exceeds slots + queue bound	5,791	EXPECTED FAILURE

Raw evidence: `ops_poc/runs/2026-10-08-recorded/` (see each check's `evidence` in `evidence/runs/2026-10-08-recorded/checks.jsonl`).

OPS-C12 · ARGUED

Autoscaling helps but does not remove provider quotas, downstream limits, budgets or tenant fairness; a queue absorbs imbalance, it does not create capacity.

Rests on: `research/sources.md`: `k8s-hpa`, `sre-overload`, `aws-queue-backlogs`, `aws-load-shedding`; OPS-E1 and OPS-E7 as illustrations, not tests of autoscaling.

OPS-C13 · ARGUED

Rollback across a stateful agent platform is harder than rolling back a stateless binary: indexes, memory, migrations and external side effects need compatible versions or reconciliation.

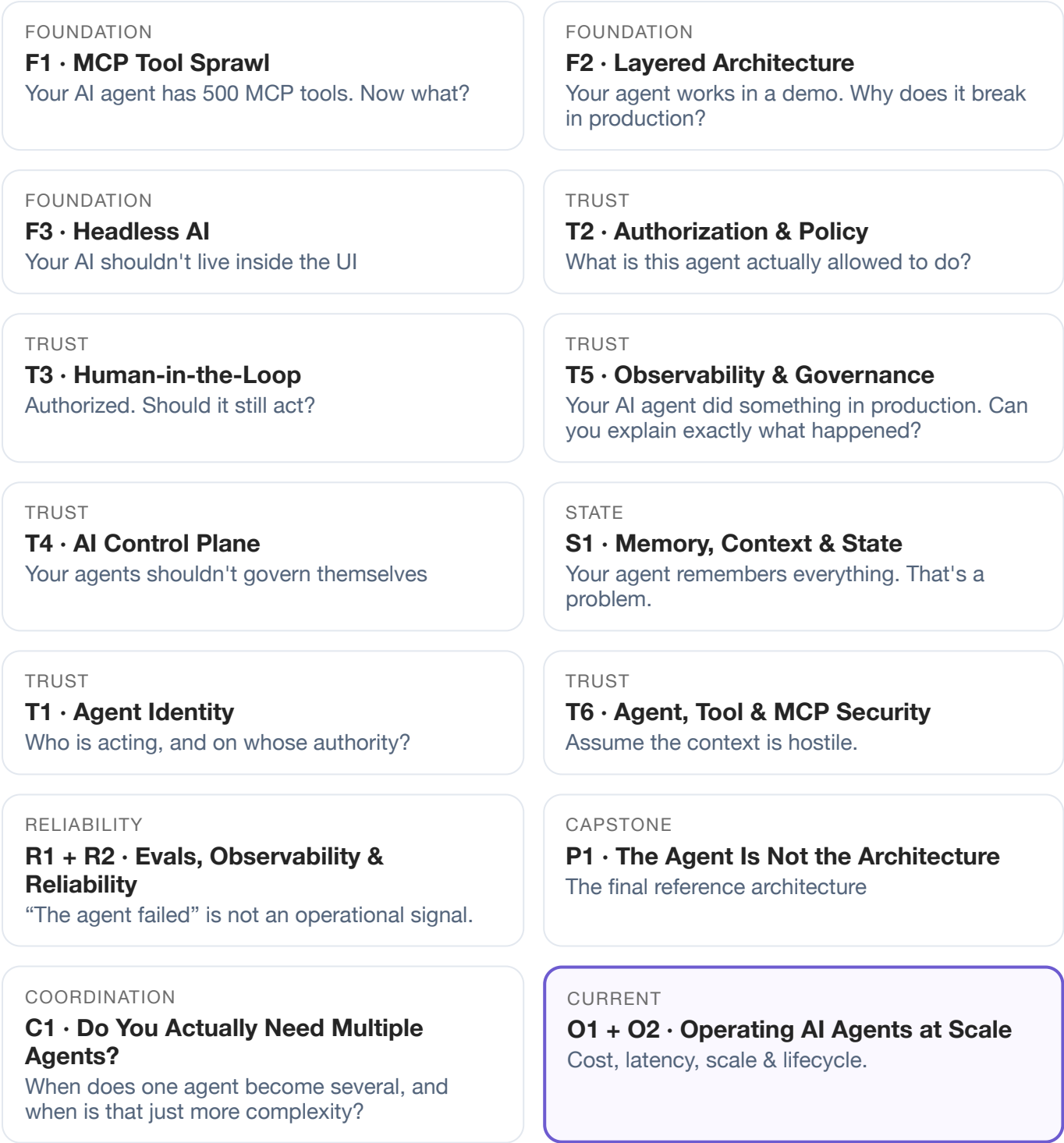
Rests on: `research/sources.md`: `aws-rollback-safety`, `sre-release-engineering`, `flagger-strategies`, `istio-mirroring`; OPS-E10-C06 shows only the side-effect half.

OPS-C14 · NOT TESTED

Real provider latency, quotas and prices; real model quality; human approval capacity; shadow traffic; real tool limits. Every number in this POC is a simulation unit.

Rests on: `ops_poc/config/*.toml` (declared parameters); `proof/manifest.toml` `[[reality]]` ARCHITECTURE.

PRODUCTION AI ENGINEERING · THE SERIES SO FAR



The full learning map (Start Here)

O1 + O2 · Production AI Engineering

Production AI Engineering · O1 + O2. Every measured number here is substituted at build time from ops_poc/runs/2026-10-08-recorded/facts.json. The platform controls are real code; the clock, providers, tools, clients and agents are a deterministic simulation in simulated milliseconds and cost units, and the whole run replays byte for byte. Figures carry their provenance.

Medium edition Technical deep dive POC README Run summary Claim evidence (JSON)
QA report